A Positive Trigenic Panel

What the last panel taught, whether the model can pick positives at all, and ten genes that test it

Michael Volk

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Abstract

The previous 010 panel was selected from predictions that were invalid, by an objective with no diversity term, on genes some of which had no training evidence. This proposes a replacement drawn from `inference_1`, the one run that survived both index defects, and it is designed against each of those three failures rather than around them.

Positive interactions are worth picking, which was not previously established. Kuzmin 2018 defines a one-sided negative call, so the earlier retrieval measurement covered negatives only. Under the symmetric Kuzmin 2020 call, $\tau > +0.08$ with $p < 0.05$, the ensemble of three checkpoints reaches precision 0.100 at $K = 100$ against a base rate of 0.526 percent, an enrichment of 19.0 times, where the additive null reaches 0.030. Average precision is 0.0352 against the null's 0.0110. At $\tau > +0.20$ the ratio widens to 0.111 against 0.0109. On held-out data the predictions are also calibrated: a regression of actual on predicted has slope 0.877, and of the 18 test records predicted above +0.20 the mean actual τ is +0.312 with two thirds being real calls.

Confidence no longer tracks absence of evidence. On the previous panel predicted interaction rose with the number of genes having no trigenic training record. Here the correlation runs the other way, Spearman +0.267 between prediction and the least-supported gene's record count.

The design is two gene-disjoint complete five-gene blocks: 10 genes, 20 triples, 20 doubles, and every gene in exactly 6 of the 20. Balance is a property of the shape rather than a penalty term, and a complete block is the cheapest closed design because τ needs all three doubles and all three singles of a triple measured beside it.

One finding constrains what can be claimed. Across 6,934,927 complete five-gene blocks, the model's positive confidence sits in a single clique: the best block away from it has a median predicted τ of 0.029, under half the training-label standard deviation of 0.063, and within the leading block the six triples containing `YER059W` score +0.590 to +0.662 while the four without it score +0.069 to +0.127. The panel is therefore built as a contrast rather than as 20 equivalent wagers, and the contrast is what makes it informative either way.

Section status: ✓ ready ■ author-reviewed, keep stable ✗ not done, agents may edit freely

Contents

1 What the last panel taught ✗

Five things went wrong with the 12-gene panel. Each one is a gate in the selection that replaces it, applied before any triple is scored rather than checked afterward.

1.1 The predictions were invalid ✗

The panel was selected from `inference_3`. Between training and that run the genome gene set went from 6,607 entries to 6,579, and the 28 absent entries are mitochondrial open reading frames that sort before every nuclear name, so every gene index shifted by exactly 28. Separately the candidate genes were never intersected with the model's gene space, and a name the index cannot resolve contributes no perturbation, so 266 of 432 stored panel combinations were scored as doubles. Rescored correctly the 20-triple build list spans -0.134 to $+0.011$ rather than the 0.409 to 0.711 it was ordered by.

`inference_1` escaped both. Its 526 genes all resolve inside the model's gene set with no alias resolution needed, so no triple collapsed, and a fresh 3,000-triple sample re-scored through the validated path reproduces the stored predictions at Pearson 0.9999983 under the 6,607-gene map. It is the only run of the three that can be used.

1.2 A hub gene took the panel over ✗

The objective maximized how many of the top k predicted triples were contained in the panel, and its tiebreak was the raw count of top-k triples containing a candidate gene, which rewards hubs directly. Nothing penalized concentration. Participation ran from 46 triples down to 16 across the 12 genes. One round earlier the same pathology collapsed `inference_2` onto a single gene, and the response was to relax the fitness thresholds rather than to change the objective.

1.3 Confidence tracked absence of evidence ✗

Predicted interaction rose with the number of panel genes having no trigenic training record: mean 0.4531 with none, 0.5300 with one, 0.6676 with two, Spearman 0.634 at $p = 1.5 \times 10^{-5}$. Genes with zero records took ranks 1 through 11 of the 39 targets. `YER079W` has no trigenic and no digenic record in this build at all, because the query enables only the two Kuzmin trigenic datasets.

1.4 Two genes could not have been built as named ✗

`YLR312C-B` and `SPH1` occupied separate wells on the same plate and are one locus: R64 records `YLR313C` with both as aliases. Twelve strain labels were eleven distinct loci. `YLR312C-B` is additionally an SGD merged open reading frame with no discrete protein product. Two further panel names, `YJL017W` and `YKL200C`, are stale systematic names whose current forms are `YJL016W` and `YKL201C`, and both scored wrong for that reason alone.

1.5 A fitness filter was defeated by its own aggregation ✗

`YJR060W`, CBF1, entered the panel despite a viability floor of 0.90 because single-mutant fitness was aggregated across strain backgrounds with `max()`. Its KanMX strain measures 0.923 and its NatMX strain 0.590; the mean, 0.757, would have excluded it. The same aggregation let 8 of 66 panel doubles report a queried double-mutant fitness below 0.90, and the one cross that failed to construct contains it.

1.6 The cost driver is τ closure, not construction ✗

A trigenic score consumes seven fitnesses,

$$\tau_{abc} \;=\; f_{abc} - f_{ab}f_c - f_{ac}f_b - f_{bc}f_a + 2f_af_bf_c, \tag{1}$$

so every triple needs all three of its doubles and all three of its singles measured beside it. A set cover over the previous 20 triples needs only 7 doubles to build them, but closure needs 33, and the plate carried 65 strains. Building to the cover rather than to the closure yields 20 fitness values and no interaction at all. The same-batch rule follows: τ is a difference, so a shared day deviation cancels only when every order sits on the same plates, and a carried-forward single-mutant reference costs up to 1.46 times the standard error.

1.7 The claim must be epistasis, not a ladder ✗

The previous round framed its claim as an ascending fitness ladder, wild type below single below double below triple. That is not reachable on a deletion panel scored on growth. None of the 12 singles measured above wild type, the genome-wide deletion ceiling is 1.1118, only 2.7 percent of singles are significantly above wild type, and in Kuzmin the fraction above wild type *falls* with order, 23.0 to 13.8 to 6.9 percent. The claim that remains available is positive epistasis: the triple grows better than the expectation built from its own singles and doubles, whether or not it beats wild type.

2 Whether the model can pick positives ✗

Nominating positive interactions had not been tested. The earlier retrieval measurement used the Kuzmin 2018 call, which is one-sided by construction: "we used an established interaction magnitude cut-off for digenic interactions ($p < 0.05, |\varepsilon| > 0.08$) and trigenic interactions ($p < 0.05, \tau < -0.08$)", with positives excluded because "We focused exclusively on the analysis of deleterious negative trigenic interactions." The symmetric form belongs to Kuzmin 2020 and the 2021 protocol, which do score positives, and it is what makes the question askable on the same records.

2.1 Positives are well represented in the labels ✗

Of the 376,732 records in this build, 20,426 have $\tau > +0.08$ and 1,773 have $\tau > +0.20$, against a label standard deviation of 0.0633 and a maximum of 1.128. The model trains on them.

2.2 Retrieval, and the margin is wider than on negatives ✗

Precision at $K = 100$ on the held-out test split, 37,673 records.

Model	P@100	enrichment	average precision
Kuzmin 2020 call, $\tau > +0.08$ and $p < 0.05$, base rate 0.526%			
B1 additive ridge	0.030	5.7×	0.0110
B5 nonlinear MLP	0.040	7.6×	0.0166
CGT ensemble	0.100	19.0×	0.0352
magnitude only, $\tau > +0.20$, base rate 0.422%			
B1 additive ridge	0.040	9.5×	0.0109
B5 nonlinear MLP	0.100	23.7×	0.0377
CGT ensemble	0.210	49.8×	0.1110

Table 1: Retrieving published positive trigenic interactions on the random-over-records test split. The transformer's average precision is 3.2 times the additive null's at the published-style call and 10.2 times at $\tau > +0.20$. On the negative side the same ratio was 1.9.

The margin over a model that cannot represent interaction is proportionally larger on positives than on negatives, which is the opposite of what the one-sided published call would suggest.

2.3 Large predictions are calibrated ✗

The previous panel's collapsed doubles scored near 0.71, more than ten label standard deviations above the mean, and that inflation was the signature of the bug. A top prediction on `inference_1` is +0.690, so the same check is owed here.

Binning the held-out test predictions against actual τ: the 18 records predicted above +0.20 have mean actual τ of +0.312 and 66.7 percent are real calls at $\tau > +0.08$; the 5 predicted above +0.30 have mean actual τ of +0.802 and all are real. A regression of actual on predicted has slope 0.877 and intercept −0.0018. Test-split predictions themselves span −0.576 to +0.631, so a prediction near +0.65 is inside the range the model demonstrably emits on labeled data rather than outside it.

2.4 Confidence now tracks presence of evidence ✗

Over the 200 most positive `inference_1` triples, the Spearman correlation between prediction and the least-supported gene's trigenic record count is +0.267 at $p = 1.3 \times 10^{-4}$, and between prediction and the mean record count +0.279. Thirty-one of the 200 contain a gene with no record, not the whole list. The direction is the reverse of the previous panel's.

The panel genes back this at the label level. `YER059W` appears in 1,097 training triples of which 16.3 percent exceed +0.08, a 3.0 times enrichment over the 5.4 percent base rate. `YKR028W` reaches 14.7 percent. The model's preference is visible in the labels it was trained on.

2.5 What none of this covers ✗

Every number above comes from the split that is random over records, where a Kuzmin query double appears in training and test at once. On a query-pair disjoint split the additive null falls from 0.400 to 0.127 ± 0.033 test Pearson and the nonlinear baseline drops below it. The transformer has never been refit there. Only 22 of the 420 query doubles have both members inside the 526-gene space, so `inference_1` is very nearly query-pair disjoint by construction and its nominations live in exactly the regime that has not been measured.

The positive side of that split is underpowered rather than clearly collapsed. It holds 51 published positive calls in 37,680 records, a base rate of 0.135 percent, and the additive ridge retrieves 3 of them in its top 100. The enrichment ratio that produces, 22.2 times, is not comparable to the random split's and rests on three records. Negative retrieval on the same split falls to 1.8 times, which is close to chance, but the positive comparison cannot yet be called either way.

3 The design ✗

3.1 Complete blocks ✗

A complete design on k genes contributes $\binom{k}{3}$ triples using $\binom{k}{2}$ doubles, and every gene appears in exactly $\binom{k-1}{2}$ of them. Three consequences follow, and together they answer the previous panel's failures without a single penalty term.

Balance is structural. At $k = 5$ every gene sits in 6 of the block's 10 triples, so no objective can produce a hub because the shape forbids one.

Closure is cheapest. Twenty triples chosen freely from an 8-gene set pull in most of $\binom{8}{2} = 28$ doubles; two complete 5-blocks give 20 triples on exactly 20 doubles. A single complete 6-block is cheaper still, 20 triples on 15 doubles, but it spans only 6 genes and offers no internal contrast.

Two blocks are two experiments. A block that fails at the bench does not take the round with it, and a disjoint second block tests whether the ranking means anything away from the first.

3.2 The objective is the worst checkpoint ✗

Two training runs share only 0.04 to 0.14 of their top 100 on this space, so a mean across checkpoints can be carried by one optimistic run. Blocks are therefore ranked on the median of their ten triples' *minimum* across the three checkpoints, which makes disagreement cost score rather than average out. The median is used rather than the block's own worst triple because the worst is set by whichever single triple the checkpoints disagree on most, which for the leading block is the one triple lacking its hub gene.

3.3 Gates applied before scoring ✗

Every gene must resolve through the shared reconciler to a current systematic name, that name must be in the model's 6,607-gene set, no two panel genes may resolve to the same locus, and the gene must carry at least 200 trigenic training records. The support floor is set where the previous panel's distribution cliff was, from 519 down to 10 down to 0. Single-mutant fitness is aggregated across strain backgrounds by mean, and the per-strain spread is reported rather than collapsed.

A pair is treated as constructible exactly when some triple in the inference space contains it, which inherits the upstream guarantee: every `inference_1` triple is a viable three-clique whose singles all exceed 0.9 fitness and whose three doubles were all measured at 1.0 or above.

3.4 Cost ✗

	this panel	previous panel
genes	10	11
triples	20	20
doubles for τ closure	20	33
singles	10	11
strains on the plate	51	65
triples per gene	6 for every gene	46 down to 16

Table 2: The complete-block design closes τ on 20 triples with 13 fewer doubles and 14 fewer strains, and holds every gene at the same participation.

Fifty-one strains at 384 wells is 7 replicates each with wild type on every plate. Colonies within a plate are pseudo-replicates, so replicate count is not the lever; on clean plates the between-plate standard deviation is 0.028 and the colony term dominates. The strain-count cost of a larger panel has not been re-planned since that correction, and the two figures in the record disagree, +27 percent against under 10 percent for a comparable panel growth. That is worth settling before the layout is fixed.

4 The panel ✗

Ten genes, all viable, all supported.

Block	Gene	single-mutant fitness	trigenic records	% $\tau > +0.08$
1	`YER059W`	1.024	1097	16.3
1	`YGL060W`	1.016	1321	3.6
1	`YGR121C`	1.004	1413	9.6
1	`YHR003C`	0.987	1356	6.6
1	`YPL181W`	1.036	295	3.4
2	`YDR423C`	0.994	1049	9.6
2	`YHR204W`	1.049	288	4.2
2	`YJR082C`	0.976	796	2.0
2	`YKR028W`	0.971	1323	14.7
2	`YLR258W`	0.989	1114	5.1

Table 3: Costanzo single-mutant fitness aggregated by mean across strain backgrounds. No gene shows any disagreement between its KanMX and NatMX strains, which is the check that `YJR060W` failed on the previous panel. The base rate for $\tau > +0.08$ across the whole build is 5.4 percent.

4.1 Three tiers, and the contrast is the point ✗

Tier	Triples	median predicted τ	range	median spread
A, contains `YER059W`	6	+0.635	+0.590 to +0.662	0.081
B, block 1 without `YER059W`	4	+0.099	+0.069 to +0.127	0.446
C, disjoint control	10	+0.029	+0.011 to +0.044	0.017

Table 4: Predicted τ is the worst of three checkpoints; spread is the range across them. All 20 triples are positive under all three checkpoints. The training-label standard deviation is 0.0633.

The tiers are not three guesses of decreasing quality. They are a designed contrast, and the reason for it is a property of the model rather than of the selection. Across 6,934,927 complete five-gene blocks in this space the positive confidence is concentrated in one clique: the best block sharing no gene with it has a median predicted τ of 0.029, under half the label standard deviation, and no sixth gene extends the leading block above a median of 0.024. The distribution puts that in scale: the median block scores -0.0005 and the 99.99th percentile 0.0196, so the leading block's 0.607 sits 31 times above the highest percentile the rest of the space reaches. Within that block the six triples containing `YER059W` score $+0.590$ to $+0.662$ and the four without it score $+0.069$ to $+0.127$.

So the model's positive claim is close to a single-gene claim, and the panel is built to find that out. If measured τ orders A above B above C, the model is ranking trigenic interactions. If A is high while B and C are indistinguishable, it has learned one gene's marginal behavior and not interaction structure. If none of the three separates, the ranking carries nothing at this scale. Every outcome is readable, which a panel drawn only from tier A would not be.

Tier B is where the checkpoints disagree most, spreads of 0.34 to 0.51 against 0.08 in tier A and 0.02 in tier C. Removing `YER059W` from a gene set the model is confident about does not lower its prediction so much as destabilize it.

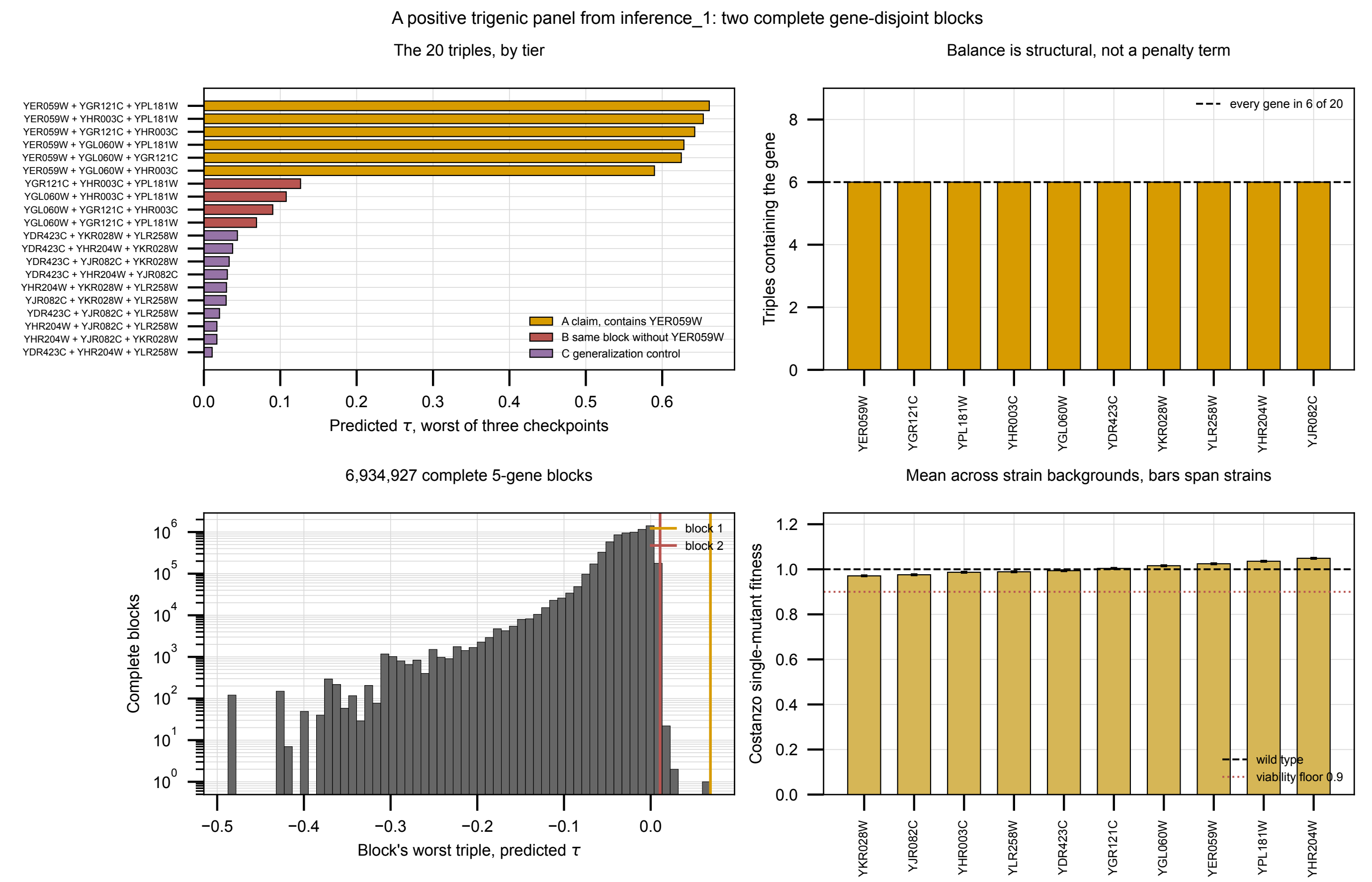

Figure 1: The panel. Bars are the worst of three checkpoints, colored by tier. Every gene sits in exactly 6 of the 20 triples. Both chosen blocks sit at the extreme right of the distribution over all complete five-gene blocks. Every gene is within 3 percent of wild-type single-mutant fitness. Generated by `experiments/010-kuzmin-tmi/scripts/positive_panel_selection.py`.

5 Restricting to strong interactions

Requiring the model to call an interaction *strong* rather than merely positive shrinks the pool by two orders of magnitude and removes the diversity the panel was built around.

5.1 Where the threshold comes from

Kuzmin's ordinary call is a conjunction at $|\cdot| > 0.08$ with $p < 0.05$. Baryshnikova 2010 adds a stringent tier that is sign-asymmetric, $\varepsilon < -0.12$ negative and $\varepsilon > 0.16$ positive, and the positive arm is the strong cut used here. It is a digenic threshold applied to a trigenic score, which is an approximation rather than a published standard: no stringent tier for positive *trigenic* interactions exists.

5.2 How many ✗

Predicted τ cut	by ensemble mean	every checkpoint above	distinct genes
$> +0.08$	209	39	23
$> +0.12$	74	24	14
$> +0.16$ *(strong)*	**39**	**15**	**10**
$> +0.20$	29	13	8
$> +0.30$	18	11	7
$> +0.50$	8	7	6

Table 5: Triples of the 4,370,595 in `inference_1` clearing each cut. The two columns differ by 2.6 times at the strong cut because a mean can be carried by one optimistic checkpoint. Requiring all three is the rule a build list should use.

So the answer is **15 triples**, over 10 genes, under the conservative rule. Two fail the training-support gate, `YGR193C` at 10 records and `YHL003C` at 0, which leaves **13 triples over 8 genes**.

5.3 What it costs and what it removes ✗

	strong only	two-block design
triples	13	20
genes	8	10
doubles for τ closure	19	20
strains on the plate	41	51
triples containing `YER059W`	12 of 13	6 of 20
control arm	none	10 triples

Table 6: The strong restriction is 10 strains cheaper and loses both the generalization control and the balance.

The concentration becomes total. Twelve of the 13 gated strong triples contain `YER059W`, and the single exception is `YBR001C + YHR003C + YPL181W` at +0.481. Every gene participation count is set by that one gene: no complete block survives the cut, so the design reverts to a ranked list.

5.4 The tier is retrievable, at low absolute precision ✗

Model	P@100	enrichment	average precision
$\tau > +0.16$ *and* $p < 0.05$, *base rate 0.181%*			
B1 additive ridge	0.030	16.6×	0.0069
B5 nonlinear MLP	0.020	11.1×	0.0081
CGT ensemble	0.070	38.8×	0.0300

Table 7: Retrieval at the strong tier on the random-over-records test split. The transformer's average precision is 4.3 times the additive null's, wider than the 3.2 at the ordinary call, but only 7 of its 100 most positive predictions are real strong calls.

The margin over the additive null widens as the tier tightens, 1.9 times at the negative call, 3.2 at the ordinary positive call, 4.3 here and 7.2 at $\tau > +0.20$. Absolute precision falls the other way, so a strong-only list is a higher-confidence ranking of a much rarer event, not a more reliable one.

On the query-pair-disjoint split the strong tier holds 13 positives in 37,680 records and both baselines retrieve none of them in their top 1,000. That comparison is not evidence of collapse, it is an absence of measurable signal at this rarity, and the transformer is unmeasured there either way.

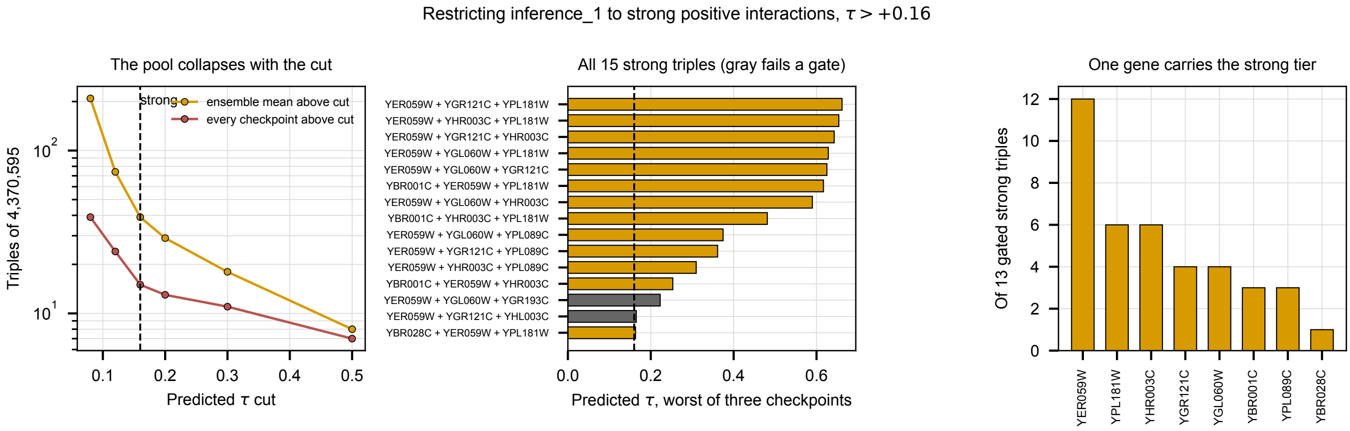

Figure 2: The strong pool. Gray bars fail a gene gate. Generated by `experiments/010-kuzmin-tmi/scripts/strong_positive_candidates.py`.

5.5 The choice this forces ✗

Building only strong predictions buys a higher bar on 13 triples and gives up the contrast that made the 20-triple design readable. If tier A of that design measures high, the strong-only list would have measured high too, and neither would say whether the model ranks interactions or one gene. Keeping four non-`YER059W` triples from the same block and ten from a disjoint block costs 10 strains and is what separates those two outcomes.

6 Established, and not ✗

6.1 Established ✗

The model retrieves published positive trigenic interactions well above base rate on the held-out split, and its margin over a model that cannot represent interaction is proportionally wider there than on negatives: average precision 0.0352 against 0.0110 at the Kuzmin 2020 call, and 0.111 against 0.0109 at $\tau > +0.20$.

Predictions of the size this panel rests on are calibrated on labeled data. Records predicted above +0.20 have mean actual τ of +0.312, the regression of actual on predicted has slope 0.877, and the model emits the same range on the test split that it emits here.

Confidence tracks the presence of training evidence rather than its absence, reversing the previous panel's relationship, and the leading gene's positive enrichment is visible in the labels at 3.0 times base rate.

The design closes τ on 20 triples with 20 doubles and 51 strains against the previous 33 and 65, and holds every gene at 6 of 20 triples by construction.

6.2 Not established ✗

Whether the ranking survives a disjoint split. Every retrieval and calibration number here comes from a split that is random over records. The additive null falls from 0.400 to 0.127 ± 0.033 once query doubles are made disjoint, and the transformer has not been refit there. Since only 22 of the 420 query doubles are representable in the 526-gene space, this panel sits in that unmeasured regime almost entirely.

Whether tier A is an interaction claim or a gene claim. That is the question the panel is built to answer, and it is open until it is measured. `YER059W` is enriched for strong interactions in both directions, 16.3 percent above +0.08 and 14.1 percent below −0.08, so a large measured effect of either sign is consistent with it being a high-variance gene rather than with the specific triples being right.

Whether the assay can report τ at all yet. On the previous round singles reproduced Costanzo fitness at $r = 0.706$ but doubles did not ($r = 0.255$), and measured digenic ε against published ε gave $r = -0.245$. The diagnosis was a normalization artifact: the denominator rests on a single wild-type colony, 8 of 12 singles scored above on-plate wild type, and all 13 ε came out negative with $\mathrm{corr}(\varepsilon, f_a f_b) = -0.926$. That shift is undiagnosed, and no correction should be applied before it is.

The strain-count cost of this panel. The between-plate standard deviation of 0.140 that the current layout advice rests on was an artifact of one failed plate; clean plates agree to 0.028, which inverts the advice from “plates are the only lever” to “the colony term dominates”. The design has not been re-planned since, and the record carries two incompatible figures for what packing more strains costs.

6.3 Next ✗

1. Re-run the layout design at the corrected between-plate standard deviation before fixing the plate, and settle the strain-count cost.
2. Fix the wild-type normalization, since τ closure is worthless if ε is not scale-invariant.
3. Train the transformer on the query-pair-disjoint split. Every claim here is conditional on a split whose leakage is quantified and whose replacement has not been run.
4. Read the panel from the plating artifact each run rather than freezing a gene set as a constant, which is how the previous basis silently went stale.
5. Record every cross attempt with a date and an outcome, including failures.